Finn – Covenant Check Agent: Summary

Case study submission · Tharu Antony Melath · November 2026

What I built

Finn is an email agent written in Python. It watches a Gmail inbox, receives a request with the business plan (`Wirtschaftsplan_2027.xlsx`) attached, checks the financial covenants of the four loan contracts, and replies in the same thread. The reply contains a summary, a table (plan value, threshold, headroom, met yes/no), the source of every number (contract § and page; Excel sheet and cell) and the full calculation path. A JSON audit trail is attached.

How it works

1. **Contracts:** I analysed the four contracts with AI assistance (Claude), checked the results myself, and recorded each covenant's threshold, definition, rounding rule and verbatim quotes in a covenant register (`covenants.json`). On every run, Finn reads the register, locates each stored quote in the contract PDF and determines the page number automatically. If a quote is no longer found (e.g. because the contract changed), Finn adds a warning to the reply.
2. **Business plan:** the Excel file is read with cell references; totals are recalculated in code, and the plan is checked for consistency (balance sheet balances, loan schedule matches balance sheet and P&L).
3. **Calculation:** each covenant is calculated exactly as its own contract defines it, including the traps: Contract 1 has no covenants; Contract 2 counts 50 % of the Sonderposten as equity; Contract 3 leaves out the special repayment and the pension unwinding; Contract 4 excludes the subordinated shareholder loan.
4. **Reply:** an HTML email built from the calculated results.

Why the calculations are done in code, not by the LLM

All ratios are calculated in Python with exact decimal arithmetic and the contracts' own rounding rules. LLM output is not reproducible: the same input can give slightly different numbers or interpretations on different runs, and LLMs can also make arithmetic mistakes. In covenant checks, that can decide between “met” and “breached”. For example, Contract 4's leverage is 3.37 (met) if the subordinated shareholder loan is excluded correctly, but 3.57 (breach) if it is not. Code gives the same result every time and can be tested and audited.

AI was used to read the contracts; I checked the result and fixed it in the covenant register. At runtime the LLM is optional and only handles language: understanding the request and writing the summary text. A guard rejects any LLM text that contains a number not produced by the code. Without an LLM, Finn uses text templates and still works fully.

Result Plan 2027

All covenants are met. The Konsortialkredit is tight on both covenants: equity ratio 30.7 % vs. ≥ 30.0 %, and leverage 3.37x vs. ≤ 3.50x.

Testing

19 automated tests cover the covenant values, the contract traps, rounding, faulty Excel files and the full email flow. I also tested it live via Gmail.

Next steps with more time

Automatic detection and LLM-assisted proposals for contract changes (with human approval), scenario analysis (“what if EBITDA −10 %?”), and running in the cloud instead of on a laptop.